

STUDY GUIDE

TACET — Cross-Question Defense Manual (SIH26151)

[cite: 1]

Ordered by critical severity. Every response reflects the live submitted build — never upgrade claims under questioning[cite: 1].

Three Non-Negotiable Rules Before You Start:[cite: 1]

1. **Admit gaps before you are caught.** A limit stated calmly reads as technical competence[cite: 1]. The same limit extracted under pressure reads as a bluff[cite: 1].
2. **Never claim a real person is a criminal.** All forum personas are strictly fictional[cite: 1]. Imputing guilt to real individuals invites immediate failure upon artifact inspection[cite: 1].
3. **Say "lead", never "identification".** The platform outputs weighted leads for manual review; it never automates accusations[cite: 1].

TIER 1 — QUESTIONS THAT CAN END YOUR RUN[CITE: 1]

Q1. "Where did you get the dark web data? Show me the dumps." [cite: 1]

WHY THEY ASK[CITE: 1]

Tests whether illegal live scraping occurred or evidence was fabricated[cite: 1].

SAY THIS[CITE: 1]

"We don't ship real dark web dumps, and deliberately won't claim to[cite: 1]. The forum personas and posts are fictional—written so we never republish victim data or assert claims against real people[cite: 1]. Everything touching the real world comes from authoritative records: 373 OFAC-designated wallets from the US Treasury SDN list, a real multi-author corpus from 20 Newsgroups, and live chain lookups via blockchain.info[cite: 1]. The ingest pipeline is format-agnostic (JSON, CSV, BBCode, raw text), allowing authorized agencies to plug in lawful evidence dumps unchanged[cite: 1]."

IF PRESSED: "SO YOU HAVE NO DARK WEB DATA AT ALL?" [CITE: 1]

"Correct, by deliberate engineering design[cite: 1]. Handling actual dumps requires law enforcement chain-of-custody credentials we cannot hold as students[cite: 1]. We built and verified the analysis layer that parses lawful data[cite: 1]."

Q2. "Is this legal? Aren't you de-anonymizing people without consent?" [cite: 1]

SAY THIS[CITE: 1]

"The engine never reaches out to the Tor network, runs zero live surveillance, and operates strictly on lawfully retained agency evidence[cite: 1]. In India, this aligns with the IT Act and DPDP Act provisions[cite: 1]. Generated case dossiers incorporate §65B / Bharatiya Sakshya Adhiniyam 2023 certificate language following the NIST SP 800-86 workflow[cite: 1]. Access is strictly role-gated with hash-chained operational logs[cite: 1]. We supply the investigative capability; legal authority stays with the agency[cite: 1]."

Do not say: "It's fine because it's public data." [cite: 1] Tor dumps contain stolen personal data; asserting this invites fatal procedural follow-ups[cite: 1].

Q3. "This could be used to stalk or harass innocent people. What stops that?"[cite: 1]**SAY THIS[CITE: 1]**

"Like any digital forensics suite, the software itself cannot physically block operator misuse[cite: 1]. What we implement is deep accountability: role-based privilege controls, an append-only hash-chained audit log, non-verdict confidence tiering, and zero automated intervention triggers[cite: 1]. Systems run entirely air-gapped on agency premises[cite: 1]. Misuse is checked by auditing and chain governance[cite: 1]."

Q4. "What happens if your system is wrong and someone gets accused?"[cite: 1]**SAY THIS[CITE: 1]**

"We engineered against that hazard[cite: 1]. Match confidence is never presented as definitive proof, but divided into analytical bands: >75% 'Likely Same Actor', >55% 'Probable Link', >35% 'Weak', and <35% 'Insufficient'[cite: 1]. Analysts receive an explicit score breakdown separating deterministic artifacts (wallets, PGP) from probabilistic stylistic signals[cite: 1]. Verification is mandatory before operational pivot[cite: 1]."

TIER 2 — SCIENTIFIC & STYLOMETRIC SCRUTINY[CITE: 1]**Q5. "You trained on 20 Newsgroups. Forum posts are short, slangy, and multilingual. Your 86% doesn't transfer."[cite: 1]****SAY THIS[CITE: 1]**

"You are correct, and we make no claim of zero-shot domain transfer[cite: 1]. 86.2% is a cross-validated benchmark over an established corpus (751 documents, 20 distinct authors, stripped of quoting, headers, and signatures)[cite: 1]. Stylistic confidence decays drastically over short, noisy posts[cite: 1]. For that exact reason, style accounts for only 36% of our heuristic score, while hard deterministic artifacts account for 42%[cite: 1]. In production, retraining on agency-held data is required[cite: 1]."

Q6. "Stylometry is junk science. Authors change style deliberately."[cite: 1]**SAY THIS[CITE: 1]**

"Style alone is brittle—we treat it as a secondary, low-confidence pivot[cite: 1]. Adversarial evasion like translation loops and LLM rephrasing breaks text metrics[cite: 1]. Hard cryptographic artifacts like PGP keys and Bitcoin addresses cannot be stylistically paraphrased, which is why they dominate our scoring weights[cite: 1]."

Q7. "Is your model actually trained, or hard-coded?"[cite: 1]**SAY THIS[CITE: 1]**

"Inspect the ML panel: it references the live training artifact—86.2% 5-fold CV accuracy (± 0.8 SD), 82.8% held-out test score, 0.79 Macro-F1 across 89,401 TF-IDF features[cite: 1]. Running ``python server/ml/train.py`` executes full model re-convergence[cite: 1]."

FOLLOW-UP: "YOUR PRELIMINARY RUN SCORED 99%. WHY IS IT 86% NOW?"[CITE: 1]

"Our baseline run reached 99.4% due to feature leakage—the model read the ``From:`` header lines[cite: 1]. We sanitized headers, quoted blocks, and signatures, dropping performance to a realistic 86%[cite: 1]. Showing the 99% would represent benchmark contamination[cite: 1]."

Q8. "Why not use RoBERTa, as the problem statement requests?"[cite: 1]**SAY THIS[CITE: 1]**

"We deliberately substituted the classification engine for practical forensic considerations[cite: 1]. TF-IDF paired with Logistic Regression executes fully deterministic inferences, runs air-gapped without external compute dependencies, and offers explainable feature coefficients[cite: 1]. Deep LLMs introduce non-determinism and massive computational load[cite: 1]. Because scoring consumes an abstract vector interface, dropping in fine-tuned RoBERTa embeddings is an isolated phase-two upgrade[cite: 1]."

Q9. "Your UI states forum proximity is 10% of the score. Is it implemented?"[cite: 1]**SAY THIS (NEVER BLUFF)[CITE: 1]**

"It is defined in config but marked 'reserved' in the interface[cite: 1]. The active scoring weights are 42% artifacts, 36% stylometry, and 12% temporal pattern matching (normalizing to roughly 47/40/13 across the 0.90 envelope)[cite: 1]. Full proximity scoring requires reconstructing parent-child thread trees from unstructured dumps, which is currently scheduled for subsequent pipeline iterations[cite: 1]."

TIER 3 — CRYPTO & ON-CHAIN ANALYSIS[CITE: 1]**Q11. "Do you actually do blockchain analysis?"[cite: 1]****SAY THIS[CITE: 1]**

"We perform structural validation (Base58Check and bech32 parsing), correlate address reuse across identity nodes, match entities against 373 OFAC SDN records, and query live chain balances[cite: 1]. We do **not** implement UTXO peeling graph analysis or change-address clustering heuristics[cite: 1]."

Q12. "Show me it works on a real wallet."[cite: 1]**EXECUTION[CITE: 1]**

"Query HYDRA Market's sanctioned wallet (OFAC SDN Entry 36216, CYBER2)[cite: 1]. Live lookups return 1,459 confirmed transactions and over 13,519 BTC in cumulative flow, triggering sanctions tags[cite: 1]."

TIER 4 — EVIDENCE INTEGRITY & SYSTEMS[CITE: 1]**Q15. "How do I know your evidence wasn't tampered with?"[cite: 1]****SAY THIS[CITE: 1]**

"Every post and ingested address stores a SHA-256 hash[cite: 1]. Actions append to an internal hash-chain log where every record cryptographically binds the previous link[cite: 1]. This makes retroactive tampering readily detectable, not mathematically impossible—an attacker with direct database access could rewrite the chain, which is why external ledger timestamping remains the planned hardening step[cite: 1]."

Q19. "The problem statement specifies Neo4j. Where is it?"[cite: 1]**SAY THIS[CITE: 1]**

"We avoided a heavy Neo4j deployment to guarantee that forensic workstations can run self-contained, air-gapped instances via embedded SQLite[cite: 1]. The schema uses typed relational node and edge representations that directly mirror a property graph[cite: 1]. Scaling to multi-million edge corpora would justify swapping in a remote Neo4j instance[cite: 1]."

CRITICAL DEFENSE TABLES

Strict Red Lines — Never Utter These Phrases[cite: 1]

Never Say[cite: 1]	Why It Ends Badly[cite: 1]
"We scraped Breached/Dread/Exploit." [cite: 1]	Admits unauthorized network intrusion and unlawful data acquisition[cite: 1].
"Target X is a real threat actor." [cite: 1]	Creates severe legal/defamation exposure; persona links are simulated[cite: 1].
"We fully implemented RoBERTa." [cite: 1]	Untrue; breaks immediately upon inspection of model checkpoints[cite: 1].
"The chain is 100% immutable." [cite: 1]	Locally stored hashes can be rewritten if the database file is compromised[cite: 1].
"Processes 50,000 posts per second." [cite: 1]	Unsubstantiated claim that collapses under real benchmark tests[cite: 1].
"Our dark web accuracy is 86%." [cite: 1]	The 86% accuracy applies exclusively to the 20 Newsgroups benchmark[cite: 1].

Live Demo Failure Drills[cite: 1]

Failure Scenario[cite: 1]	Action[cite: 1]	Statement[cite: 1]
On-chain query timeout[cite: 1]	Switch to locally cached address profile (6-hour TTL)[cite: 1].	"Displaying verified cached state; live re-query requires venue external routing[cite: 1]."
Server failure / crash[cite: 1]	Restart process and issue `POST /api/ingest/seed` [cite: 1].	"The test data corpus is fully deterministic; re-seeding takes under 2 seconds[cite: 1]."
Network visual tangled[cite: 1]	Click Reset View ; isolate node types[cite: 1].	"Applying category scoping filters to clear high-degree visual clutter[cite: 1]."

The 4 Core Axioms to Memorize[cite: 1]

1. **Data Origin:** Personas are mock data to prevent real-world harm; sanctions, blockchain structures, and baseline NLP corpora are fully authoritative[cite: 1].
2. **Model Rigor:** 86% is a cross-validated benchmark on formal text, not a dark-web performance claim—this is why deterministic artifacts carry 42% weight against 36% for style[cite: 1].
3. **Chain Integrity:** Hashes provide local tamper-evidence; true immutability requires external ledger anchoring[cite: 1].
4. **Scope:** The software produces confidence-scored investigative leads, never unsupervised conclusions[cite: 1].